Muscat 2040

Infrastructure Planning - Executive Summary

Rihal CODESTACKER 2026
Data Analytics Challenge
Ali Al Zaabi

The Challenge

Muscat Governorate's population has grown from 632,073 in 2003 to an estimated 1.46 million in 2023 - a compound annual growth rate of 4.3%. As Oman Vision 2040 drives economic diversification and mega-project development, Muscat's population trajectory will determine whether its healthcare and education infrastructure can keep pace with demand.

This analysis models **three population scenarios through 2040** and quantifies the resulting infrastructure gaps in hospital beds and schools.

Key Findings

Population Projections

Scenario	2040 Population	Growth from 2023	Assumptions
Base Case	~2.59 million	+1.14 million	3.0% natural growth + 0.5% migration
High Growth	~3.33 million	+1.87 million	4.0% growth + 1.0% migration (Vision 2040 boom)
Low Growth	~2.05 million	+590,000	2.0% growth + 0% net migration

The base case (~2.6M) is calibrated against official planning targets: the Ministry of Housing projects Muscat at ~2.5M by 2040, and NCSI demographic projections imply a range of 2.5-2.7M. The high growth scenario represents a full Vision 2040 boom with unconstrained expatriate inflow.

Healthcare: A Significant and Widening Gap

> Muscat currently has **~2,692 hospital beds** (35% of Oman's 7,691 total) - about **1.8 beds per 1,000** population. This is already **40% below the WHO benchmark** of 3.0 beds per 1,000.

BASE CASE SHORTFALL	HIGH GROWTH SHORTFALL	LOW GROWTH SHORTFALL
5,090	**7,300**	**3,450**
beds needed (10 hospitals)	beds needed (15 hospitals)	beds needed (7 hospitals)

Even the most conservative projection requires **seven new major hospital facilities**. This is the most urgent infrastructure gap.

Education: Sustained Construction Needed

With 18% of Muscat's population in the school-age bracket (5-18 years), current capacity of ~317 schools at an average of 615 students each is under pressure.

- **Base Case:** ~759 schools needed by 2040 - **442 additional schools**, requiring ~26 new schools per year.
- **High Growth:** ~975 schools needed - 658 additional, or ~39 per year.
- **Low Growth:** ~599 schools needed - 282 additional, or ~17 per year.

Expansion should prioritize fast-growing corridors: **Seeb, Bawshar, and Al Amerat**, where residential development is outpacing school construction.

Recommended Actions

1. **Phase hospital expansion immediately.** Begin with a new 500-bed facility in the Bawshar-Seeb corridor by 2030, with a second phase in Al Amerat by 2033. The current ratio of 1.8 beds per 1,000 is 40% below the WHO benchmark - this is already overdue.
2. **Launch a school construction program of 17-26 new schools per year.** Partner with private sector education providers to accelerate capacity, particularly in areas of new residential development. Prioritize schools in wilayats where population density is growing fastest.
3. **Establish a population monitoring dashboard for infrastructure planning.** Real-time demographic tracking - linked to civil registration, visa data, and municipal records - will allow planners to adjust construction pipelines 3-5 years ahead of demand rather than reacting to overcrowding.

Analysis prepared for Rihal CODESTACKER 2026 Data Analytics Challenge. Full methodology and reproduction instructions in Technical Appendix.

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Annex - Data Sources & References

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Official Government & Statistical Sources

Source	Data Used	Link
NCSI (National Centre for Statistics and Information)	Population data, projections 2020-2040, population clock	ncsi.gov.om
Oman Data Portal	Hospital beds in Muscat, health statistics	data.gov.om
Ministry of Health (MOH)	92 hospitals, 7,691 beds, 14.9 beds per 10,000	MOH Annual Health Report 2023
Ministry of Education (MOE)	1,268 schools, 782,818 students (2023/2024 & 2025/2026)	MOE Statistical Yearbook
Oman Vision 2040 Implementation Unit	Official Vision 2040 document and targets	oman2040.om
Ministry of Housing & Urban Planning	Urban Strategy 2040 features, Muscat 2040 target ~2.5M	housing.gov.om
Supreme Planning Framework (SPF)	Oman Vision 2040 full report	spf.gov.om
Oman News Agency	Official government news releases on population, health, education	omannews.gov.om

International Organizations

Source	Data Used	Link
WHO Global Health Observatory	Hospital beds per 1,000 benchmark (3.0 standard)	who.int/data/gho
WHO EMRO Region Report	GCC average healthcare capacity (~2.2 per 1,000)	WHO Eastern Mediterranean Region Health Report
UNESCO Institute for Statistics	Class size and school-age population benchmarks	uis.unesco.org
UNESCWA	Oman Vision 2040 document archive	nwm.unescwa.org

Population Projections & News Sources

Source	Data Used	Link
Oman Observer	NCSI population projections 8.1-8.7M by 2040	omanobserver.om
Zawya	NCSI 2040 population projections breakdown	zawya.com
Arabian Business	Oman population forecast to 2040 (+2.4M growth)	arabianbusiness.com

Source	Data Used	Link
Times of Oman	Knowledge report on population growth by 2040	timesofoman.com
Muscat Daily	2024 population growth, Khoula Hospital expansion	muscatdaily.com
Gulf Migration (GRC)	Expatriate population data, NCSI yearbook figures	gulfmigration.grc.net

Academic & Data Platforms

Source	Data Used	Link
PMC / PubMed Central	Peer-reviewed article on Oman Vision 2040	pmc.ncbi.nlm.nih.gov
CEIC Data	NCSI population projection data series	ceicdata.com
Trading Economics	Oman crude birth rate historical data	tradingeconomics.com
City Population	Muscat governorate census data (2003, 2010, 2020)	citypopulation.de
Population Pyramids	Oman 2025 demographic structure	populationpyramids.org

Interactive Model & Repository

Live Dashboard: alizaabi.om/rihal-codestack/muscat-2040

GitHub: github.com/zaabi1995/rihal-muscat-2040 | **Technical Appendix:** `docs/technical_appendix.md`

Ali Al Zaabi - CODESTACKER 2026 Data Analytics Challenge. CAGR model with adjustable growth/migration rates. Built with Python, Streamlit, Plotly, Pandas.